

Sarah Thomas

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EDUCATION

Vellore Institute of Technology <i>B.Tech in Computer Science – Specialization in Cloud Computing and Automation</i>	September 2023 – May 2027 <i>CGPA: 8.81/10</i>
St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh <i>All India Senior School Certificate Examination</i>	March 2022 – May 2023 <i>CGPA: 7.88/10</i>
St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh <i>All India Secondary School Examination</i>	March 2020 – June 2021 <i>CGPA: 9.2/10</i>

TECHNICAL SKILLS

Cloud & DevOps: AWS(Lambda, S3, API Gateway, EC2, IAM, CloudWatch), Docker, Docker Compose, Prometheus(Metrics Telemetry), Terraform, CI/CD (via GitHub Actions), Git, GitHub, Boto3(AWS SDK)
Languages & Frameworks: Java, Python, Linux/Bash, FastAPI, REST APIs
Databases & Tools: MongoDB, SQL Server

PROJECTS

Serverless Multi-Tenant Configuration Processor

AWS Lambda, AWS S3, AWS API Gateway, Python (Boto3), AWS IAM github.com/THOMAS-SARAH/aws-serverless-config-processor

- Built an event-driven serverless microservice using AWS Lambda and Python (Boto3) to process and persist multi-tenant SaaS configuration files into AWS S3.
- Integrated AWS API Gateway HTTP endpoints with custom execution roles to manage stateless event routing and bucket write operations.
- Configured dynamic storage partitioning and structured JSON response formatting for automated tenant data management.

Containerized SaaS System Health Monitor

Docker, Docker Compose, FastAPI, Python, REST APIs, Git, GitHub https://github.com/THOMAS-SARAH/docker_health_monitor

- Engineered and containerized a REST microservice using FastAPI to monitor SaaS system availability, runtime environment parameters, and health metrics.
- Authored production-ready Dockerfiles and Docker Compose configurations to manage container networking, port mappings (8000), and dynamic environment variable injections.
- Isolated execution environments to deliver a consistent, zero-dependency deployment and testing workflow across local workstations.

AI-Driven Cloud Resource & Cost Optimizer

Prometheus, Node Exporter, Terraform, AWS, Linux/Bash <https://github.com/THOMAS-SARAH/multi-region-aws-telemetry-pipeline>

- Engineered a real-time infrastructure telemetry pipeline using Prometheus and Node Exporter to monitor system metrics, CPU/memory utilization, and workload health across cloud instances.
- Integrated Prometheus scraping endpoints with a FastAPI backend to ingest live telemetry data for automated rule-based resource optimization and anomaly detection.
- Provisioned multi-region cloud infrastructure using modular Terraform templates and built a traffic and failure simulation framework to stress-test system monitoring workflows.

CERTIFICATIONS AND EXTRA-CURRICULAR

Certifications

- ServiceNow University – Virtual Internship Program
- NPTEL – Internet Of Things, Cloud Computing and Distributed Systems
- AWS Educate – Introduction to Generative AI

Extra Curriculars

- Education Minister in School Council
- Best Girl Student of my Batch 2022-23
- Club Coordinator of E-Green Club